

Ferdia Matti Treacy

in **Ferdia Matti Treacy**

[My Portfolio Website](#)

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Education

BSc **Maynooth University**, Computer Science and Software Engineering

Sept 2023 – May 2027

- Weighted grade: **91.6% (First)** in CS department for 3rd year
- **President** of Maynooth Computer Science Society: 2026 - 2027

Employment

Intel NPU, AI SDE Intern. (*Pre silicon/Hardware Modelling team*)

Jan - Sep 2026 (full time) Sep 2026 - (part time)

- Worked as a developer on a C model used in virtual platforms by firmware and compiler teams
- Increased hardware model accuracy from **72% to 100%** in testbench alignment. Closed critical gaps between the pre-silicon model and RTL implementation.
- Profiled model with kernel blob workloads, and implemented performance fixes to **double model speed**
- Architected and deployed a **custom testing framework, now used throughout the team**, streamlining the identification of edge-case discrepancies.
- Developed **CI pipelines** with github actions for automatic builds, deployment, and uploads to the share NFS of built C models.
- **Tools Used:** C, C++, Python, SystemVerilog, Github Actions, Synopsys VCS, Intel Simics

Projects

Game Boy Advance Emulator - Low-Level Systems Project

[Web build](#) 

- Built a GBA emulator in modern C++ from first principles.
- Created an ISS for the ARM7TDMI chip, with custom ARM assembly and low-level C tests.
- Added CI pipelines for sanitizers, code coverage, profiling, and WebAssembly builds.
- **Tools Used:** C++, ARM Assembly, SDL3, CMake, WebAssembly, GitHub Actions.

Subtitle Translator - Microsoft Mentorshipped Project

[Github](#) 

- Built a full stack webapp in a agile team of 6, which was Mentored by Microsoft.
- Implemented Azure services such as containers, AI voice recognition and Live Translation.
- Built a RESTFUL API backend with expressJS and mongoDB.
- **Tools Used:** Javascript, Css, React, NodeJS, Azure Web Services, Jira, MongoDB.

ContextDeck - Durable Job Pipeline in Go, made for language capture

[Github](#) 

- Built Go backend for ingesting and processing language captures for language learners
- Implemented durable job queues, retries, and idempotent workers.
- Used PostgreSQL for state persistence and migration driven storage.
- Containerised deployment with Docker Compose for local orchestration, hosted on AWS EC2 with a basic NodeJS frontend to showcase the queue pipeline in action
- **Tools Used:** Go, PostgreSQL, Docker, SQL Migrations, AWS, Distributed Systems, NodeJS

Appendix

Third year - Highest result in Department

Computation and Complexity	99%
Computer Networks	89%
Programming Languages and Compilers	96%
Software Design	100%
Software Verification	92%
Group Project	88%

Second year - Second highest result in class

Databases	93%
Algorithms & Data Structures 1	93%
Algorithms & Data Structures 2	97%
Computer Architecture	96%
Web Information Processing	97%
Operating Sys: Communications & Concurrency	91%
Software Testing	71%
Into User Experience (UX), User Interface (UI)	74%
Software Engineering & Software Process	91%
Theory of Computation	69%
Calculus 3 (S)	78%
Introduction to Statistics (S)	91%

First year - Third highest result in class

Introduction to Computer Science I	90%
Introduction to Computer Science II	100%
Computer Systems I	88%
Computer Systems II	92%
Mechanics & Special Relativity	90%
Mechanics & Modern Physics	71%
Differential Calculus	83%
Integral Calculus	96%
Linear Algebra 1 (S)	84%